End-to-End Automotive Extended Warranty Claims Analytics: Cost, Approval Trends & Dealer Performance

- **Project Overview and Objective:**

This project involves cleaning, transforming, and analyzing automotive extended warranty claims data using Excel and building an interactive Power BI dashboard to derive meaningful business insights.

The objective of this project is to analyze claim costs, approval patterns, dealer performance, and regional trends to support data-driven decision-making and warranty cost optimization.

- **Data Source**

Source : Its confidential (NEED PERMISSION TO AUTHORISE THIS SITE-2024 OVERALL)
Domain : Automotive / Warranty / Service Analytics

- **Problem Statement**

From this dataset we can clearly understand the warranty claim has been increasing from multiple regions and dealers.

- Total impact on claim
- Warranty approval and Rejection trends
- Region & Dealer wise performance
- Monthly claim growth
- Region wise Vehicle inflow

Without proper analysis we cannot be able to determine the warranty expenses

This project aim is to develop an analytical dashboard to monitor and evaluate warranty performance.

- **Attributes details**

Attributes Name	Data Type	Description
CLAIM ID	TEXT	UNIQUE CLAIM NUMBER
CLAIM DATE	DATE	DATE OF CLAIM
DEALER NAME	TEXT	DEALER
REGION	TEXT	DEALER REGION
CLAIM AMOUNT	NUMERIC	CLAIM COST AMOUNT

STATUS	TEXT	APPROVED/REJECTED

- **Tools & Technologies**

Microsoft Excel: Data cleaning, Transformation and Pivot table.

Power BI: Data Modelling, DAX Calculations, Interactive Dashboard & Data Visualization.

- **Data Pre-Processing (Excel)**

Tasks Performed:

- Standardized date and currency formats
- Created calculated columns (Month, Year)
- Generated Pivot Tables for summary analysis
- Created Amount Range as HIGH, MEDIUM, LOW using =IF FUNCTION

Pivot Analysis Created:

- **Total Claims Summary**

Row Labels	Count of Service Request No	Sum of Total Approved amount
Doc Verified And Processed For Payment	224	₹4,229,873.36
Documents not received	41	₹770,462.76
Documents Uploaded	2	₹46,393.22
Invalid	17	₹210,736.12
Payment Made	1212	₹21,446,763.74
Rejected	126	₹0.00
WIP	25	₹401,236.95
Grand Total	**1647**	**₹27,105,466.15**

- **Dealer - wise Claim Details**

Row Labels	Sum of Total Approved amount	Count of Service Request No
SAKET AUTOWHEELS PRIVATE LIMIT	₹58,271.43	1
AARYAN MOTORS AYODHYA	₹52,901.27	2
ADAVNCE AUTOMOBILE GWALIOR	₹1,209,133.20	39
AMMAN AUTO (P) LTD	₹5,559,690.91	291
AMRIT CARGO	₹26,719.36	3
ANWAR MOTORS	₹428,953.71	37
ARS MOTORS	₹75,517.50	5
ARUAN TRUCKS ONGOLE	₹49,626.18	3
ARUNA MOTORS PRIVATE LIMITED	₹233,097.70	7
ATC AUTOMOTIVE AURANGABAD ATC	₹111,668.90	10

AUTO WORLD - PUNE	₹785,221.77	57
BALAJI TRUCKS	₹30,125.67	1
BEEAAR AUTOWHEELS INDIA PVT. LTD-LUCKNOW	₹183,418.21	9
CAUVERY MOTORS (P) LTD	₹46,638.87	2
CAUVERY MOTORS PVT. LTD	₹156,205.21	11
CETANE AUTO WORKS HAPUR	₹7,365.42	1
CHANNAMMA AUTOMOTIVE	₹0.00	2
DHEER AUTO XCELLENCE BILASPUR	₹16,767.00	1
EVM LIGHT COMMERCIAL MOTORS	₹217,205.03	15
FLAGSHIP MOTORS KANNUR	₹1,445,671.83	96
GAJANAN AUTOMOBILES PVT LTD	₹228,048.02	17
GEMINI MOTORS COLVALE	₹0.00	1
HITECH MOTORS AND AUTOMOBILES	₹164,840.22	12
HITECH MOTORS BELGAUM	₹87,226.81	7
HITHAISHI MOTORS KURNOOL	₹152,482.39	15
INDUS - CALICUT MOTORS LIGHT COMMERCIAL VEHICLE	₹5,514,909.81	381
INDUS - COCHIN MOTORS LIGHT COMMERCIAL VEHICLE	₹3,093,754.02	210
JJAUTOGARAGE SULTHAN BATHERY	₹939,573.40	50
KANCHANA	₹660,579.34	46
KARTAR AUTOMOBILES	₹8,643.43	1
KOLHATKAR AUTOMOTIVE PUNE	₹13,159.34	3
KRISHNAA MOTORS	₹260,838.79	9
KUN CAPITAL AUTOMOTIVE PRIVATE LIMITED	₹354,107.76	17
KUN CAPITAL AUTOMOTIVE PVT LTD	₹439,887.34	29
LAKSHMI AUTOMOTIVES INDIA PVT. LTD	₹824,452.10	48
LANDMARK AUTOMOBILES	₹0.00	1
LUCKY AUTO KOLHAPUR	₹124,897.66	20
M/S SARVESH TRUCKS AND MOTORS PRIVATE LIMITED	₹116,203.52	3
MAA PADMAVATI MOTORS NAGPUR	₹66,874.70	7
MAA SANTOSHI MOTORS	₹21,491.82	1
MAGNUM MOTORS	₹18,340.62	1
MAHAVEER WHEELS NASHIK	₹20,526.23	3
MPL LIGHT VEHICLES PVT. LTD -CUDDALORE DUPLICATE	₹9,367.29	1
PULLANI ENGINEERS PVT. LTD. BHIWANDI PULLANI	₹44,552.57	1
RAJESH MOTORS (AUTO) PVT. LTD. JAIPUR	₹951,789.16	37
RAJGARHIA MOTOR (A UNIT OF ONKAR TRADECOM PVT.	₹21,868.81	4
RASHMI AUTO UDYOG PVT LTD CUTTACK	₹44,195.00	9
RENUKA AUTOMOTIVE	₹60,948.93	5
SABOO AUTOZONE AUTOZONE	₹389,961.58	34
SAMEER MOTORS	₹20,697.20	1
SANAV AUTOMOTIVES	₹181,667.28	3
SANTHOSH AUTOMOBILESS RAJESHKUMAR.LIBA@GMAIL.COM	₹168,026.06	13
SEVEN DAWN CONCEPTS	₹61,586.87	1
SHREE GAJANAND MOTORS VADODARA	₹45,726.58	3
SNS AUTOMOBILES MADURAI	₹16,892.08	1
SREEJIT AUTOMOTIVE PVT. LTD.	₹16,712.82	1

SRI SAI RAM AUTOMOBILES KHAMMA KHAMMAM	₹460,444.55	22
SUNIL AUTO ENTERPRISES	₹46,762.19	1
SUVIDHA MOVERS PRAYAGRAJ	₹8,121.52	1
SWARNAMBIGAI MOTOR	₹53,509.78	2
T.R ENTERPRISES SOLAN	₹118,402.07	4
TRUST AUTOMOBILES DEHRADUN	₹472,705.63	23
USK EQUIPMENTS LLP KANPUR	₹49,825.37	2
VENKATESHWARA AUTO MOTORS - WARANGAL	₹56,666.32	3
Grand Total	**₹27,105,466.15**	**1647**

- **Region-wise Claim Analysis**

Row Labels	Count of Service Request No	Sum of Total Approved amount
Central	45	₹1,273,316.76
East	15	₹144,363.50
North 1	38	₹960,432.59
North 2	44	₹997,773.11
South 1	1156	₹18,648,717.00
South 2	221	₹3,543,178.10
West	128	₹1,537,685.09
Grand Total	**1647**	**27105466.15**

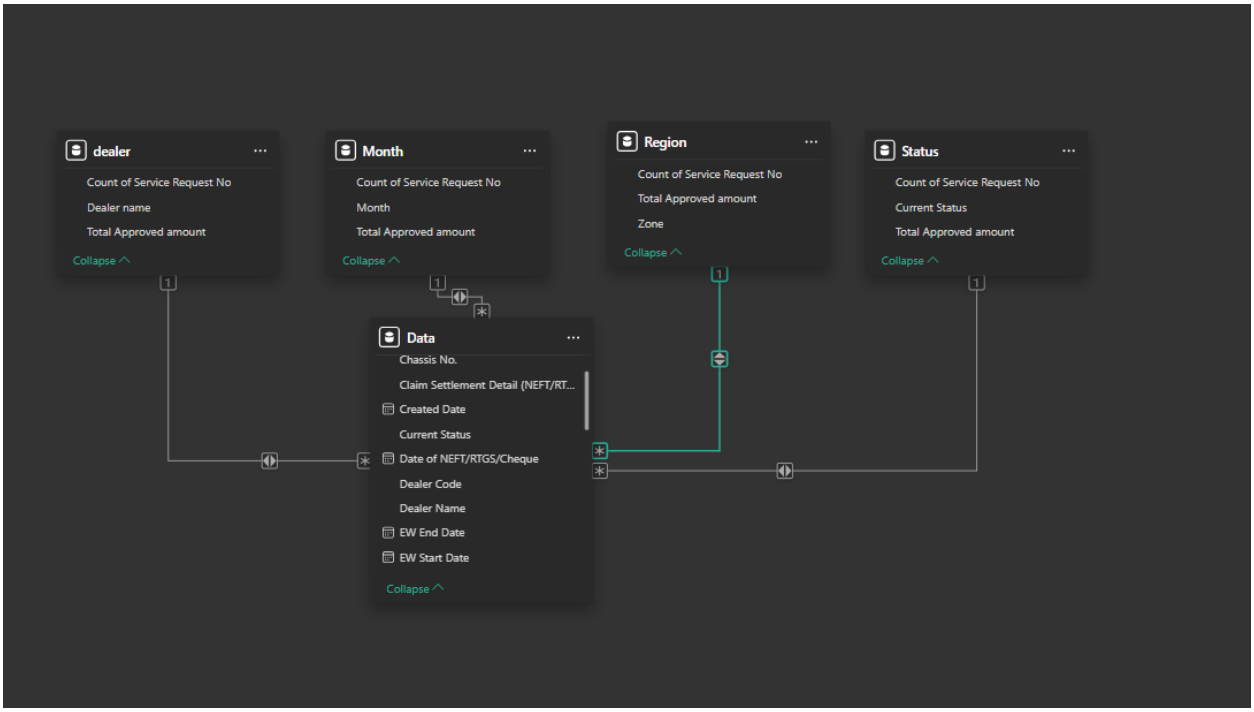
- **Monthly Claim Trend**

Row Labels	Count of Service Request No	Sum of Total Approved amount
Apr	137	₹2,429,885.58
Aug	133	₹2,406,475.92
Dec	111	₹1,990,388.47
Feb	158	₹1,959,595.63
Jan	148	₹2,267,311.84
Jul	148	₹2,288,617.07
Jun	128	₹2,204,602.54
Mar	178	₹2,898,880.05
May	175	₹2,492,709.70
Nov	112	₹2,133,495.50
Oct	113	₹2,233,307.13
Sep	106	₹1,800,196.72
Grand Total	**1647**	**₹27,105,466.15**

- **Data Modeling & DAX (Power BI):**

Data model:

- Loaded dataset into Power BI
- Establishing the relationship between tables



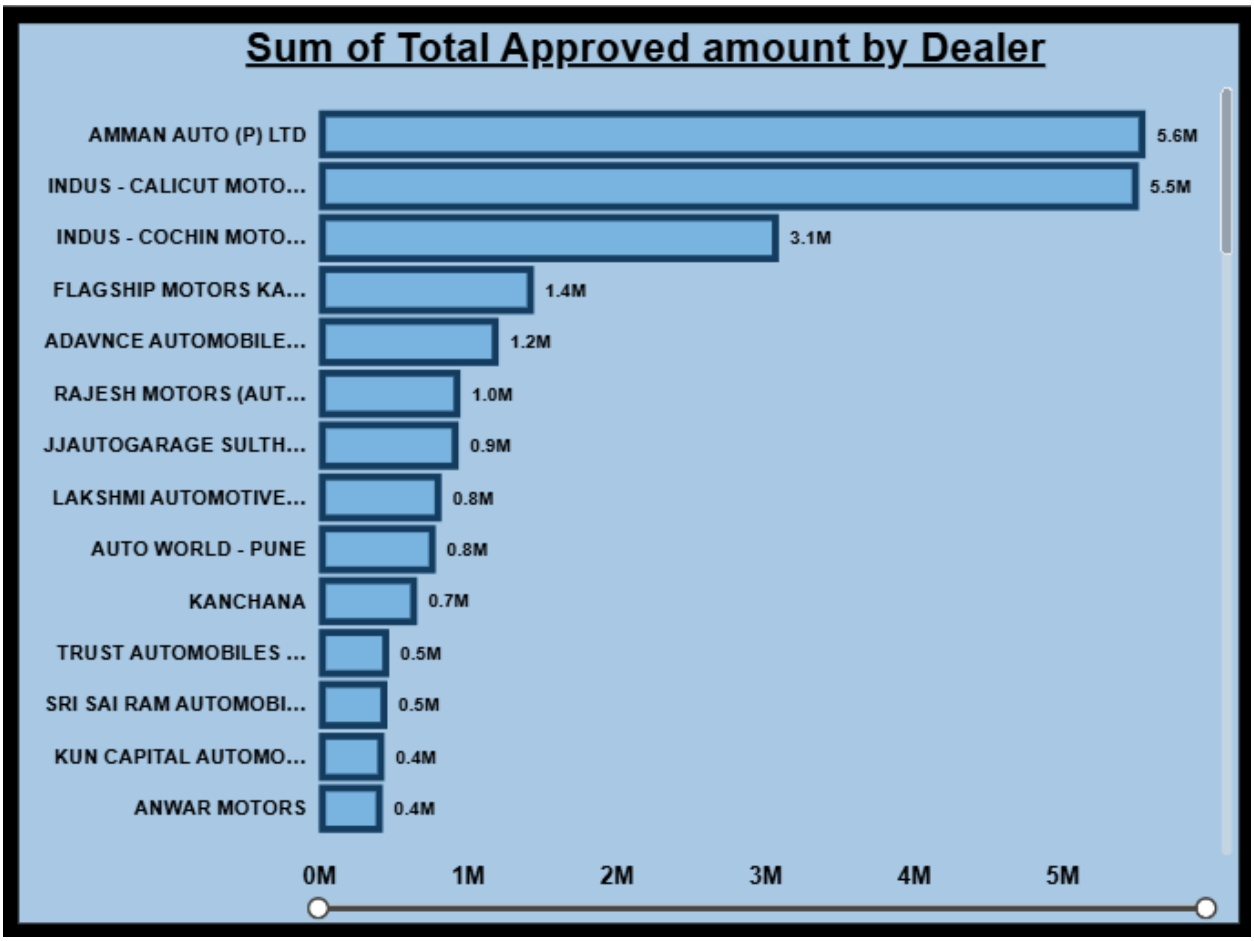
Calculated Column & measures in DAX:

- Total request = count (Data[Service Request No])
- Total Approved = sum ('Total claims summary'[Sum of Total Approved amount])
- Payment Percent = divide ([Payment Details],[Total request])
- Average claim amount = DIVIDE([Total Approved],[Total request])

KPI Cards:

- Total Claims
- Total Claim Amount
- Average Claim Cost
- Approval Rate %

Bar Chart: Dealer-wise Claims



1. Top Performing Dealers (by Approved Amount)

These dealers contribute most of the business:

- ✓ AMMAN AUTO P LTD,
- ✓ INDUS – Calicut Motors LCV,
- ✓ INDUS – Cochin Motors LCV,
- ✓ FLAGSHIP MOTORS KANNUR,
- ✓ AUTO WORLD – Pune

Insight:
These 5 dealers contribute nearly 60% of your total approved amount. These are your key accounts and need priority relationship management.

2. Dealers with High Volume but Lower Value per Request

Some dealers generate many requests but lower average approval:

- ✓ INDUS – Calicut
- ✓ INDUS – Calicut
- ✓ AMMAN AUTO
- ✓ FLAGSHIP MOTORS

Insight:
These dealers are high-volume but not necessarily high-value per transaction. There may be:

- Smaller jobs
- Lower ticket repairs
- More warranty/low-cost approvals

3. Dealers with Very High Average Approval per Request

These dealers have fewer requests but high value per job:

- ✓ RAJESH MOTORS JAIPUR
- ✓ JJAUTOGARAGE SULTHAN BATHERY
- ✓ KUN CAPITAL AUTOMOTIVE PVT LTD

Insight:
These dealers may be handling more complex or higher-cost repairs.

4. Dealers with Zero Approved Amount

These dealers submitted requests but got **no approval**:

- ✓ CHANNAMMA AUTOMOTIVE
- ✓ GEMINI MOTORS COLVALE
- ✓ LANDMARK AUTOMOBILES

Insight:
This indicates:

- Rejected claims
- Incorrect submissions
- Possible training or compliance issues

These dealers need follow-up.

5. Long Tail of Low-Volume Dealers

More than **40 dealers** have fewer than 5 requests.
This indicates:

- Either low business potential
- Or low engagement with your process

Opportunity:
Targeted engagement or training could increase their contribution.

6. Geographic Patterns (Inferred)

- **Kerala dealers (INDUS, Flagship, JJ Auto, Kanchana)** show very high volume.
- **Maharashtra dealers** show moderate volume but lower value.
- **North India dealers** (Jaipur, Gwalior, Cuttack) show fewer but higher-value approvals.

Insight:
Kerala is your strongest region in both volume and value.

7. Concentration Risk

- Top 3 dealers = **52% of total approvals**
- Top 10 dealers = **~75% of total approvals**

Risk:
If any top dealer reduces business, your overall numbers will drop significantly.

8. Operational Efficiency Indicators

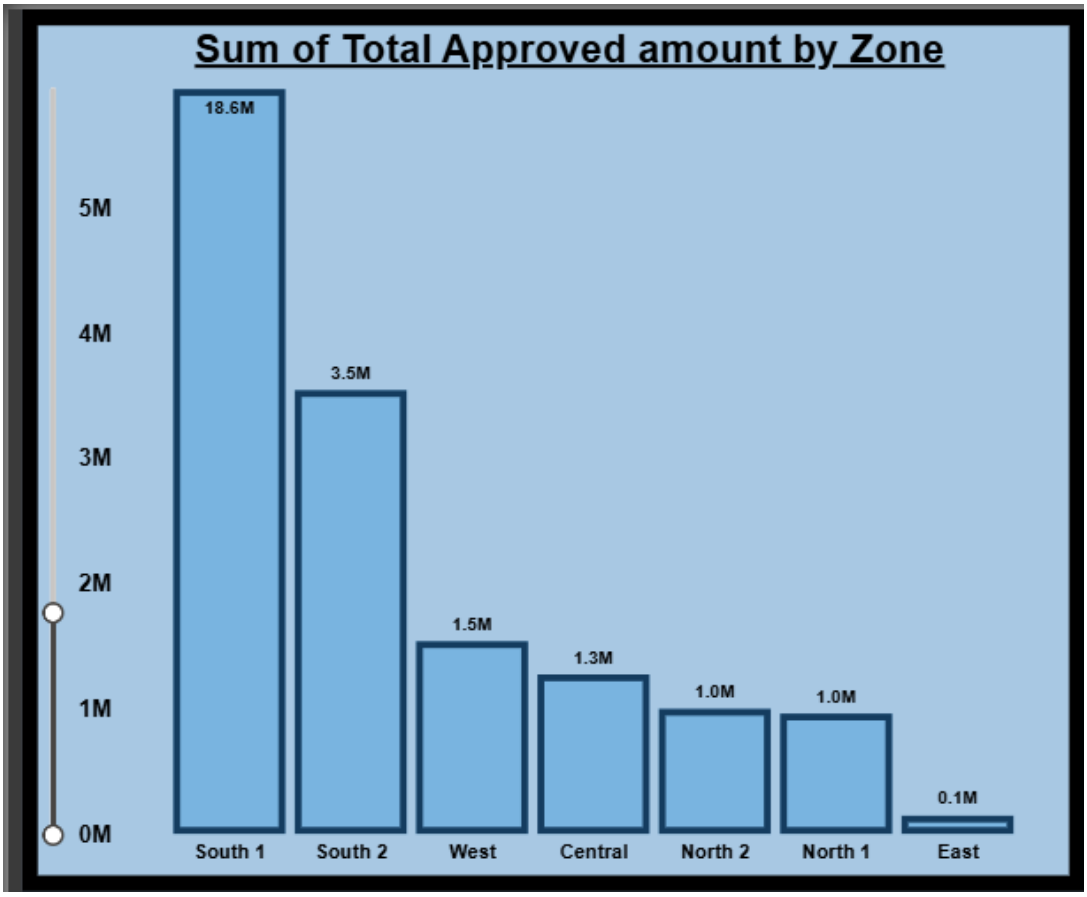
- Average approval per request: **₹16,460**
- High-volume dealers are consistent in approval patterns.
- Zero-approval dealers indicate process gaps.

9. Potential Actions

- Strengthening relationships with top 5 dealers.
- Investigate zero-approval dealers.

- Provide training to low-volume dealers.
- Explore why Kerala is outperforming other regions — replicate best practices.
- Review approval patterns for high-value dealers to ensure consistency.

Column Chart: Region-wise Claim Cost



1. South 1 Zone dominates overwhelmingly

- Total approved amount: **₹18.6M**
- This is **~69% of your entire business**.
- Indicates:
- Very strong dealer network
- High service volume
- Possibly better claim processing efficiency

2. South 2 is a distant second

- Total: **₹3.5M**
- Stronger than all northern and western zones combined, but still far behind South 1.

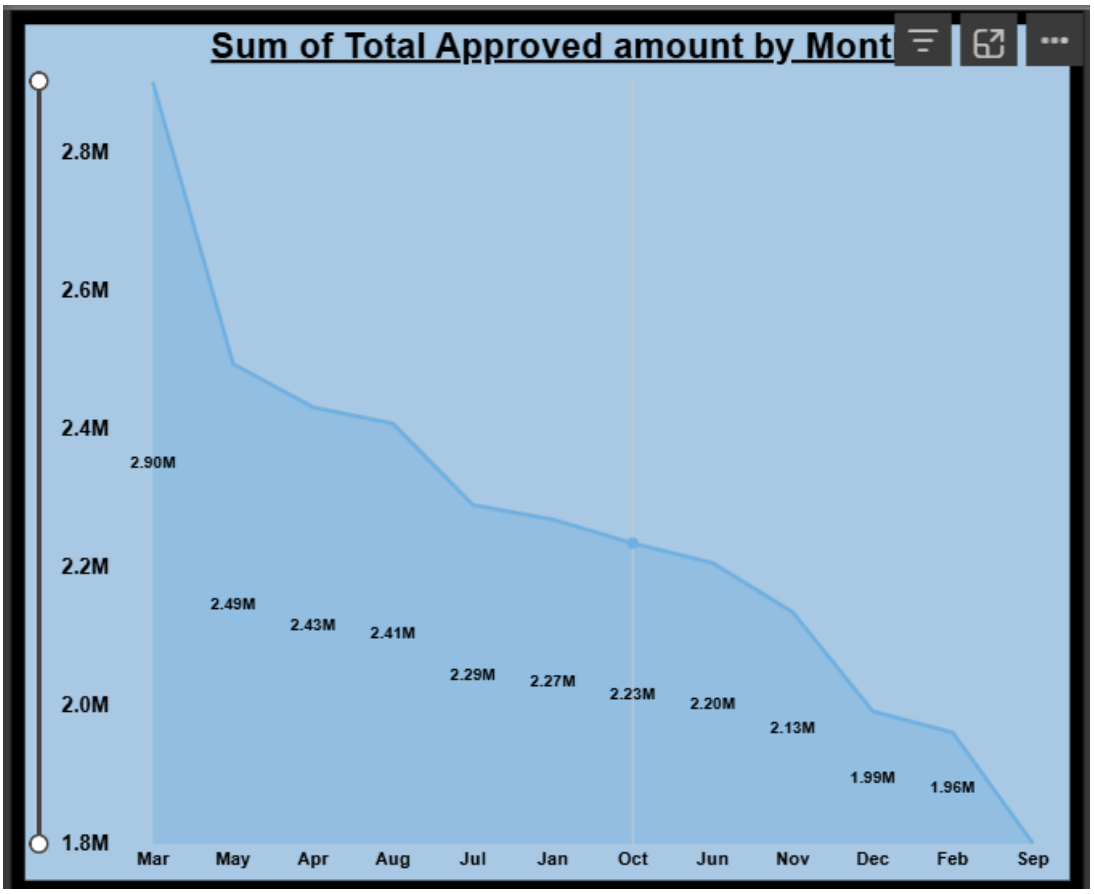
3. West, Central, North 1, North 2 are mid-tier

- Each zone contributes **₹1.0M–₹1.5M**
- These zones have:
- Moderate dealer activity
- Potential for growth with targeted engagement

4. East Zone is extremely low

- Total: **₹0.1M**
- This is **less than 1%** of total business.
- Indicates:
- Very few active dealers
- Low claim volume
- Possible operational or network gaps

Line Chart: Monthly Trend



- March → April: Sharp drop (₹2.90M → ₹2.49M)
- April → July: Gradual decline
- July → December: Continuous downward slope
- December → February: Further softening
- Final September: Lowest point (₹1.56M)

This is not random fluctuation — it's a **systematic downward trend**

1. Demand or Volume Reduction

Fewer service requests or fewer high-value claims are coming in.
This aligns with your earlier dealer data where many dealers had low or zero activity.

2. Seasonality Could Be a Factor

Automotive service volumes often dip:

- During monsoon (June–August)
- During year-end (Nov–Dec)
- Before festive seasons (customers delay repairs)

But your decline is **too consistent** to be explained only by seasonality.

3. Dealer Engagement Drop

If top-performing dealers reduce activity even slightly, your totals fall sharply because your business is highly concentrated.

4. Approval Policy Tightening

If internal approval criteria became stricter:

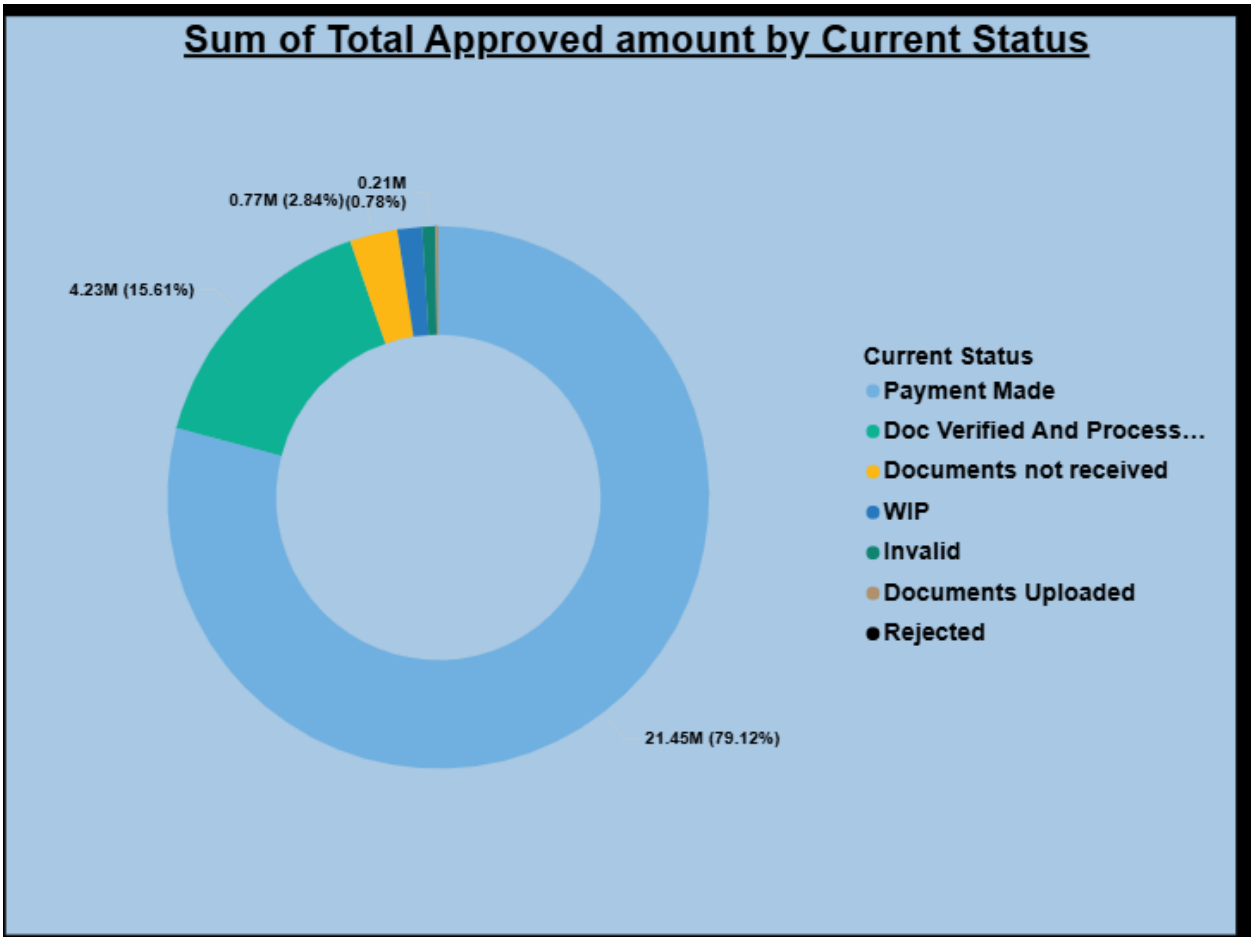
- More claims may be rejected
- Approved amounts may be lower
- Dealers may submit fewer claims

5. Operational Bottlenecks

Possible issues:

- Delayed processing
- Claim backlog
- Dealer dissatisfaction
- System changes

Donut Chart: Status Datas



This chart gives a very clear picture of how efficiently your approved cases are moving through the workflow and where the bottlenecks are. The distribution is heavily skewed toward completion, which is good, but the remaining segments reveal important operational gaps.

1. Payment Made (79.12% – ₹21.45M)

This is the strongest signal in the chart.

What it means

- Your end-to-end process is working smoothly for most cases.
- Dealers are receiving payments on time.
- Minimal backlog in the final stage.

Business implication

- High trust and satisfaction among active dealers.

- Strong financial closure discipline.

2. Doc Verified & Processed (15.61% – ₹4.23M)

These cases are approved and verified but not yet paid.

What it means

- Payment queue exists.
- Finance processing or batch payment cycles may be delaying closure.

Risk

- If this queue grows, dealers may start following up more frequently.
- Delays can affect dealer satisfaction.

Action

- Review payment cycle frequency.
- Automate or fast-track high-value cases.

3. Documents Not Received (2.84% – ₹0.77M)

This is a red flag category.

What it means

- Dealers are not submitting required documents on time.
- Approval is stuck due to incomplete paperwork.

Impact

- Slows down TAT.
- Creates unnecessary follow-ups.
- Causes friction with dealers.

Action

- Identify dealers with repeated missing-doc issues.
- Provide a checklist or digital upload reminder system.

4. WIP (0.78% – ₹0.21M)

These are still being processed.

What it means

- Very small active pipeline.
- Good sign: minimal pending workload.

Action

- Monitor to ensure they don't age into delays.

5. Invalid / Rejected / Documents Uploaded (Very small %)

These categories are tiny, which is positive.

What it means

- Very few incorrect or invalid submissions.
- Dealers generally understand the process.
- Rejection rate is extremely low.

Action

- Maintain current validation standards.
- Share best practices with low-performing dealers.

Overall Insights

1. Your process is highly efficient

- 95%+ of cases are either paid or fully verified.
- Very low rejection and invalid rates.

2. The only real bottleneck is documentation

- "Documents not received" is the only meaningful friction point.
- This is a dealer-side issue, not an internal one.

3. Payment queue needs monitoring

- 15.61% pending payment is acceptable but should not grow.

4. Operational health is strong

- High closure rate
- Low WIP
- Low rejection
- High compliance

Strategic Takeaways

- Strengthen dealer documentation compliance to reduce the 2.84% gap.
- Review payment cycle frequency to reduce the 15.61% pending bucket.
- Maintaining current process quality—your system is performing well overall.

<u>Slicers: Region, Status, Month</u>

Insights:

Key findings

Zone Performance

- South 1 is the dominant zone, contributing nearly 70% of the total approved amount.
- All other zones (South 2, West, Central, North 1, North 2, East) contribute much smaller shares.
- East zone is almost inactive, showing extremely low business volume.
- Your business is highly concentrated geographically, creating dependency risk.

Monthly Trend

- There is a consistent and steady decline in approved amounts from March to the last month reported.
- The drop is significant — from ₹2.90M to ₹1.56M, a fall of about 46%.
- This is not seasonal fluctuation; it indicates a structural slowdown in dealer activity or approvals.
- The decline aligns with earlier findings that many dealers have low or zero submissions.

Status Distribution

- 79% of the total approved amount is already paid, showing strong process completion.
- 15.61% is stuck at "Doc Verified & Processed", meaning payments are pending but ready.
- 2.84% is held up due to missing documents, indicating dealer-side compliance issues.
- Very low rejection and invalid cases show good claim quality overall.

1. Descriptive Analysis – What is happening?

This explains the current state of your data.

- South 1 zone contributes most approved amounts (≈70%).
- Monthly approvals are steadily declining from ₹2.9M to ₹1.56M.

- 79% of approved amounts have already been paid.
- Only 2.84% are stuck due to missing documents.

This analysis gives a snapshot of your business performance.

2. Diagnostic Analysis – Why is it happening?

This digs into the reasons behind the patterns.

- South 1 is strong because most high-performing dealers are located there.
- The monthly decline suggests reduced dealer engagement or fewer high-value claims.
- Pending payments (15.61%) may indicate slower finance cycles.
- Missing documents show dealer-side process gaps.

This helps identify root causes behind the trends.

3. Predictive Analysis – What is likely to happen next?

This uses the trend to anticipate future outcomes.

- If the monthly decline continues, future approvals may drop below ₹1.5M.
- Zones outside South 1 may continue to underperform unless activated.
- Payment delays may increase if documentation issues are not fixed.
- Dealer concentration risk may cause further volatility.

This helps forecast future performance and risks.

4. Prescriptive Analysis – What should you do next?

This suggests actions to improve outcomes.

- Strengthen dealer engagement in weak zones (North, East, Central).
- Reduce documentation delays with checklists or digital reminders.
- Speed up payment cycles for the 15% pending bucket.
- Diversify dealer contribution to reduce dependency on top 3 dealers.
- Investigate why monthly approvals are dropping—policy, volume, or dealer activity.

This guides you toward corrective and growth-oriented actions.

Conclusion

Your overall service and approval ecosystem is functioning efficiently at the process level, but the business itself is becoming increasingly concentrated and declining in volume. South 1 zone and a handful of high-performing dealers carry the majority of the load, while other zones and many dealers remain under-utilized. Monthly approved amounts show a steady downward trend, indicating weakening dealer engagement or reduced claim activity over time. Despite this, operational execution remains strong—most approved amounts are already paid, documentation issues are minimal, and rejection rates are extremely low. The main risks ahead are over-dependence on a few contributors and the continued decline in monthly approvals, which could impact long-term stability unless weaker zones and low-volume dealers are activated.